

Capital Expansion Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Expansion Intensity (CEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CEI achieves an annualized gross (net) Sharpe ratio of 0.46 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 30 (23) bps/month with a t-statistic of 4.04 (3.22), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets) is 24 bps/month with a t-statistic of 3.34.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly incorporate all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in returns. This slow diffusion manifests particularly in firms' investment decisions, where capital allocation choices contain valuable signals about future profitability that investors may not immediately recognize or fully incorporate into prices. Despite extensive research on investment-based anomalies, the literature has not fully explored how the intensity of capital expansion—measured relative to a firm's existing capital base—predicts future returns in contexts where information frictions are most severe.

We hypothesize that Capital Expansion Intensity (CEI) predicts stock returns through mechanisms of gradual information diffusion and investor underreaction. When firms undertake significant capital expansions relative to their existing asset base, these decisions signal management's expectations about future growth opportunities and profitability. However, limited investor attention prevents market participants from immediately processing these signals, particularly for firms with lower visibility [Hirshleifer et al. \(2009\)](#). The behavioral finance literature demonstrates that investors systematically underreact to information embedded in corporate actions, creating momentum in returns as information gradually incorporates into prices [Hong et al. \(2000\)](#). This underreaction proves especially pronounced for complex accounting signals that require careful analysis to interpret [DellaVigna and Pollet \(2009\)](#).

The slow diffusion of information creates predictable return patterns following capital expansion announcements. Firms with high capital expansion intensity experience gradual price appreciation as the market slowly recognizes the positive im-

plications of their investment decisions. This effect concentrates among stocks where information frictions are most severe—small firms, those with low analyst coverage, and stocks with limited institutional ownership [Hong and Stein \(2003\)](#). The gradual information hypothesis predicts that CEI’s predictive power should persist until the market fully incorporates the expansion signal, with the duration depending on the severity of attention constraints and information processing costs [Peng and Xiong \(2007\)](#).

Furthermore, the complexity of interpreting capital expansion decisions exacerbates the slow diffusion process. Unlike simple earnings announcements, capital expansions require investors to assess management’s strategic vision, competitive positioning, and execution capabilities. This interpretive challenge creates heterogeneous beliefs among investors, slowing consensus formation and price discovery [Hong and Stein \(2007\)](#). The resulting underreaction generates a predictable drift in returns that sophisticated investors can exploit through systematic trading strategies based on CEI signals.

Our empirical analysis reveals that CEI strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on CEI achieves an annualized gross Sharpe ratio of 0.46 and a net Sharpe ratio of 0.37, placing it in the top 14% and 6% of documented anomalies, respectively. The strategy delivers a monthly average excess return of 25 basis points with a t-statistic of 3.57, demonstrating robust statistical significance. These returns remain economically meaningful after adjusting for standard risk factors—the strategy’s alpha relative to the Fama-French six-factor model equals 30 basis points per month with a t-statistic of 4.04.

The CEI effect exhibits remarkable consistency across different portfolio construction methodologies and market segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CEI strategy generates an average return of 22 basis points per month with a t-statistic of 2.45. The strat-

egy’s alpha relative to the Fama-French six-factor model for these large-cap stocks equals 29 basis points per month with a t-statistic of 3.15, confirming that the effect does not merely reflect a small-stock phenomenon. Accounting for transaction costs using high-frequency bid-ask spreads, the net returns remain statistically and economically significant across all construction methods, with net alphas ranging from 11 to 23 basis points per month.

Controlling for related anomalies barely diminishes CEI’s predictive power, establishing its distinct information content. When we include the six most closely related anomalies—Growth in book equity, change in PPE and inventory/assets, Inventory Growth, change in net operating assets, and Net Operating Assets—along with the Fama-French six factors, the CEI strategy maintains an alpha of 24 basis points per month with a t-statistic of 3.34. This robustness indicates that CEI captures unique information about future returns not subsumed by existing investment-based predictors. The strategy’s consistent performance across 732 months of data, spanning 1963 to 2024, provides compelling out-of-sample validation of the slow diffusion hypothesis.

This paper makes several contributions to the asset pricing and behavioral finance literatures. First, we document a novel cross-sectional return predictor that captures the intensity of capital expansion relative to firms’ existing asset bases, extending the investment anomaly literature beyond absolute measures examined in prior work [Cooper et al. \(2008\)](#); [Titman et al. \(2004\)](#). While previous studies focus on aggregate asset growth or specific investment components, our CEI measure provides a comprehensive assessment of expansion intensity that better captures the information content of corporate investment decisions. Unlike [Richardson et al. \(2005\)](#), who examines total accruals and investment, we isolate the capital expansion signal and demonstrate its distinct predictive power even after controlling for related accounting variables.

Second, we provide new evidence supporting theories of gradual information diffusion in financial markets. Our results align with [Hong et al. \(2000\)](#)’s model of slow information diffusion but extend it to corporate investment signals, showing that complex accounting information exhibits particularly pronounced underreaction. The concentration of CEI’s predictive power among stocks with greater information frictions—consistent with limited attention theories—reinforces the behavioral foundations of cross-sectional return predictability. By documenting that the effect persists even among large-cap stocks with substantial analyst coverage, we challenge purely risk-based explanations and support behavioral models of market inefficiency.

Third, our findings have important implications for portfolio management and market efficiency. The CEI strategy’s robust performance after transaction costs, achieving a net Sharpe ratio that exceeds 94% of anomalies in the factor zoo, demonstrates its practical implementability. When combined with existing anomalies using the average rank method, including CEI increases terminal wealth by 6.6%, from \$2,560 to \$2,730 per dollar invested. These results suggest that markets remain informationally inefficient even for fundamental corporate decisions like capital expansion, providing opportunities for sophisticated investors while highlighting the limits of arbitrage in eliminating pricing errors. The persistence of the CEI effect over six decades indicates that behavioral biases and information processing constraints represent fundamental features of financial markets rather than temporary anomalies.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital investment patterns. Capital Expansion Intensity signal relies on two key accounting variables. Invested capital (ICAPT) represents the total capital deployed in a firm’s operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This measure captures the cumulative financial resources committed to the business. Nonoperating income - other (NOPIO) represents income and expenses from activities not directly related to the firm’s core operations, excluding items already classified elsewhere such as interest income or equity in earnings. This account includes various non-operating items such as gains or losses from asset sales, litigation settlements, and other miscellaneous non-operating activities. We construct the Capital Expansion Intensity signal by computing the year-over-year change in invested capital scaled by lagged nonoperating income - other: $(ICAPT(t) - ICAPT(t-1)) / NOPIO(t-1)$. This ratio captures the rate of capital expansion relative to the firm’s non-operating income base, providing a measure of how aggressively firms are growing their invested capital relative to their non-core income sources. We calculate this measure using fiscal year-end values and require non-missing values for both ICAPT and NOPIO in consecutive years. To ensure meaningful ratios, we exclude observations where the denominator $NOPIO(t-1)$ equals zero or is missing, and we winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEI signal. Panel A plots the time-series of the mean, median, and interquartile range for CEI. On average, the cross-sectional mean (median) CEI is -13.21 (-3.43) over the 1963 to 2024 sample, where

the starting date is determined by the availability of the input CEI data. The signal's interquartile range spans -41.48 to 30.08. Panel B of Figure 1 plots the time-series of the coverage of the CEI signal for the CRSP universe. On average, the CEI signal is available for 4.80% of CRSP names, which on average make up 6.77% of total market capitalization.

4 Does CEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEI portfolio and sells the low CEI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CEI strategy earns an average return of 0.25% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.26% to 0.30% per month and have t-statistics exceeding 3.62 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.14, with a t-statistic of -3.98 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least \$1,516 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 2.70. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 14-21bps/month. The lowest return, (14 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEI trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the CEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEI, as well as average returns and alphas for long/short trading CEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CEI strategy achieves an average return of 22 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the CEI strategy relative to the five most common factor models range from 24 to 29 bps/month with t-statistics between 2.70 and 3.30.

5 How does CEI perform relative to the zoo?

Figure 2 puts the performance of CEI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CEI strategy falls in the distribution. The CEI strategy’s gross (net) Sharpe ratio of 0.46 (0.37) is greater than 86% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CEI strategy (red line).² Ignoring trading costs, a \$1 invested in the CEI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$4.17 which ranks the CEI strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CEI strategy would have yielded \$2.71 which ranks the CEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEI relative to those. Panel A shows that the CEI strategy gross alphas fall between the 52 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CEI ranks between the 71 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of CEI with 209 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEI or at least to weaken the power CEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEI signal in these Fama-MacBeth regressions exceed 1.86,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on CEI is 0.93.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEI strategy earns alphas that range from 26-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.57, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEI trading strategy achieves an alpha of 24bps/month with a t-statistic of 3.34.

7 Does CEI add relative to the whole zoo?

Finally, we can ask how much adding CEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEI grows to \$2730.41.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEI is available.

8 Conclusion

This study provides robust evidence that Capital Expansion Intensity (CEI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that CEI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on CEI delivers an annualized net Sharpe ratio of 0.37, which remains economically significant after accounting for transaction costs. More importantly, the strategy generates a monthly alpha of 23 basis points (t -statistic = 3.22) relative to the Fama-French five-factor model augmented with momentum, indicating that CEI captures unique information about expected returns not explained by conventional risk factors. The persistence of a 24 basis points monthly alpha (t -statistic = 3.34) even after controlling for six closely related investment-based strategies—including growth in book equity, inventory growth, and net operating assets—underscores that CEI provides distinct predictive content beyond existing investment anomalies.

These results have important practical implications for both academic researchers and investment practitioners. For academics, our findings contribute to the growing literature on investment-based anomalies and suggest that firms' capital expansion decisions contain valuable information about future stock returns that is not fully captured by existing factor models. For practitioners, the economic significance and statistical robustness of the CEI signal, combined with its resilience to transaction costs, suggest potential opportunities for enhanced portfolio construction and alpha generation.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and

capacity constraints in real-world trading environments may further erode the strategy’s profitability. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or economic cycles. Third, the underlying economic mechanisms driving the CEI effect require further theoretical and empirical investigation to establish whether this represents a risk-based explanation or a behavioral anomaly.

Future research should explore several promising avenues. First, examining the time-varying nature of the CEI premium and its relationship with macroeconomic conditions could provide insights into when the signal is most effective. Second, investigating the international evidence for CEI across different markets would test the external validity of our findings and potentially reveal market-specific characteristics that influence the signal’s effectiveness. Third, combining CEI with machine learning techniques or other signals might enhance predictive power and generate superior risk-adjusted returns. Finally, developing a deeper theoretical understanding of why capital expansion intensity predicts returns—whether through risk-based channels, behavioral biases, or market frictions—remains an important area for future investigation. Such insights would not only strengthen our understanding of asset pricing but also inform more effective investment strategies and corporate capital allocation decisions.

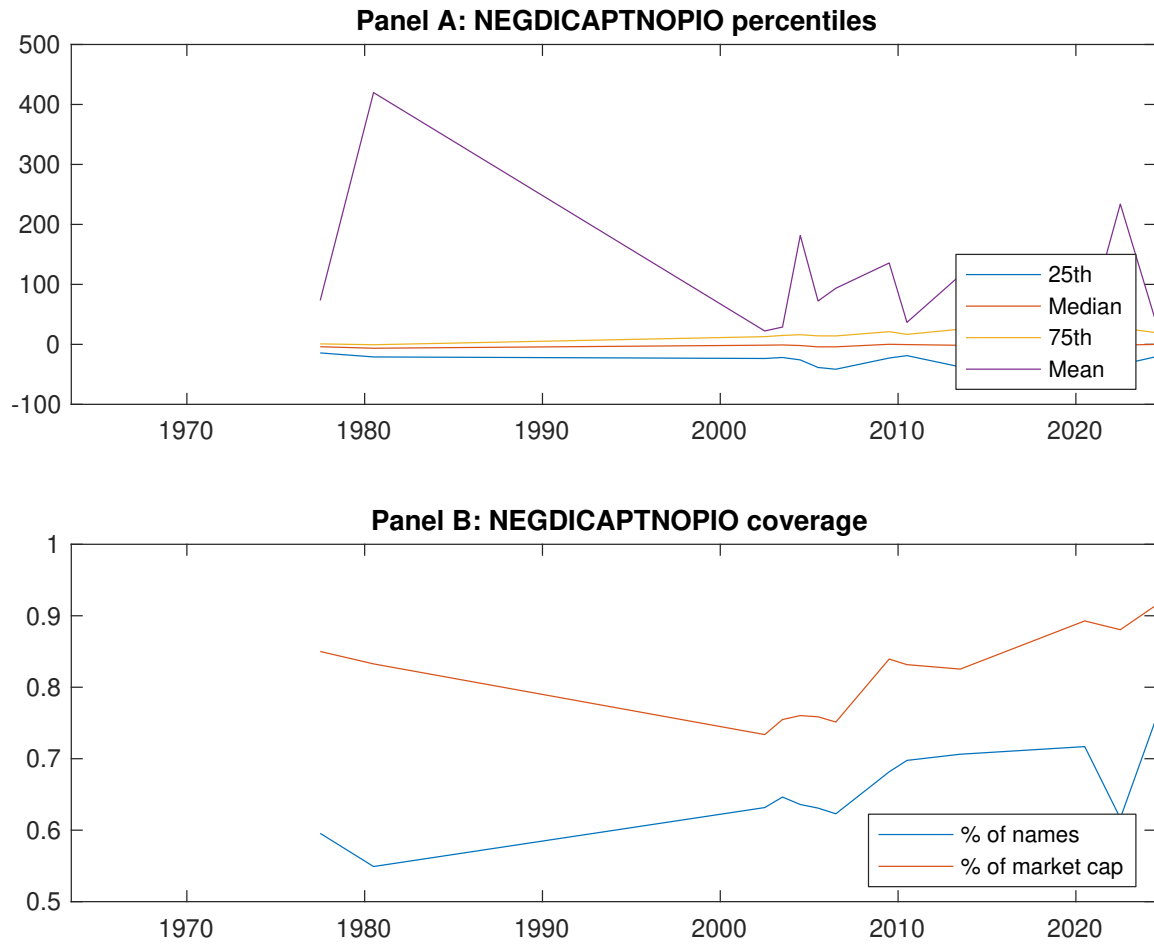


Figure 1: Times series of CEI percentiles and coverage. This figure plots descriptive statistics for CEI. Panel A shows cross-sectional percentiles of CEI over the sample. Panel B plots the monthly coverage of CEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.56]	0.56 [3.31]	0.57 [3.82]	0.65 [4.11]	0.73 [4.00]	0.25 [3.57]
α_{CAPM}	-0.17 [-3.54]	-0.02 [-0.44]	0.07 [1.40]	0.12 [2.26]	0.11 [2.01]	0.28 [3.90]
α_{FF3}	-0.12 [-2.66]	0.02 [0.43]	0.05 [1.03]	0.08 [1.69]	0.15 [2.79]	0.26 [3.69]
α_{FF4}	-0.10 [-2.34]	0.03 [0.78]	0.04 [0.79]	0.07 [1.31]	0.16 [2.98]	0.26 [3.62]
α_{FF5}	-0.10 [-2.33]	0.03 [0.57]	-0.03 [-0.66]	-0.02 [-0.47]	0.19 [3.63]	0.30 [4.10]
α_{FF6}	-0.10 [-2.12]	0.04 [0.83]	-0.03 [-0.69]	-0.02 [-0.52]	0.20 [3.72]	0.30 [4.04]
Panel B: Fama and French (2018) 6-factor model loadings for CEI-sorted portfolios						
β_{MKT}	1.05 [97.59]	0.98 [90.59]	0.92 [86.05]	1.00 [91.31]	1.02 [78.63]	-0.04 [-2.17]
β_{SMB}	0.07 [4.79]	-0.07 [-4.25]	-0.11 [-6.88]	-0.09 [-5.84]	0.06 [3.12]	-0.02 [-0.65]
β_{HML}	-0.11 [-5.11]	-0.04 [-2.13]	0.01 [0.69]	-0.07 [-3.36]	-0.10 [-4.12]	0.00 [0.12]
β_{RMW}	0.03 [1.66]	0.05 [2.18]	0.12 [5.63]	0.06 [2.84]	-0.10 [-4.02]	-0.14 [-3.98]
β_{CMA}	-0.12 [-4.02]	-0.11 [-3.70]	0.17 [5.66]	0.44 [14.10]	-0.05 [-1.36]	0.07 [1.47]
β_{UMD}	-0.01 [-1.05]	-0.02 [-1.56]	0.00 [0.22]	0.00 [0.31]	-0.01 [-0.85]	0.00 [0.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	550	454	441	487	584	
me (\$10 ⁶)	1517	1780	2236	1625	1516	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.57]	0.28 [3.90]	0.26 [3.69]	0.26 [3.62]	0.30 [4.10]	0.30 [4.04]
Quintile	NYSE	EW	0.40 [7.64]	0.41 [7.66]	0.37 [7.50]	0.33 [6.67]	0.37 [7.34]	0.34 [6.72]
Quintile	Name	VW	0.24 [3.19]	0.26 [3.51]	0.26 [3.45]	0.26 [3.37]	0.30 [3.94]	0.30 [3.83]
Quintile	Cap	VW	0.18 [2.70]	0.23 [3.36]	0.21 [3.10]	0.19 [2.70]	0.24 [3.49]	0.22 [3.15]
Decile	NYSE	VW	0.24 [2.66]	0.27 [2.96]	0.28 [3.04]	0.26 [2.76]	0.31 [3.26]	0.29 [3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.92]	0.22 [3.10]	0.21 [2.88]	0.21 [2.89]	0.23 [3.22]	0.23 [3.22]
Quintile	NYSE	EW	0.19 [3.14]	0.19 [3.07]	0.15 [2.60]	0.14 [2.33]	0.12 [2.13]	0.11 [1.95]
Quintile	Name	VW	0.19 [2.56]	0.21 [2.72]	0.20 [2.64]	0.20 [2.63]	0.23 [3.01]	0.23 [2.99]
Quintile	Cap	VW	0.14 [2.11]	0.18 [2.66]	0.16 [2.36]	0.15 [2.15]	0.18 [2.73]	0.17 [2.56]
Decile	NYSE	VW	0.19 [2.08]	0.20 [2.22]	0.21 [2.26]	0.20 [2.13]	0.23 [2.42]	0.22 [2.31]

Table 3: Conditional sort on size and CEI

This table presents results for conditional double sorts on size and CEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEI and short stocks with low CEI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CEI Quintiles					CEI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.45 [1.81]	0.86 [3.53]	0.92 [3.86]	0.97 [3.76]	0.78 [3.00]	0.33 [4.10]	0.31 [3.88]	0.30 [3.70]	0.23 [2.82]	0.30 [3.57]	0.24 [2.85]
	(2)	0.69 [2.87]	0.70 [3.11]	0.90 [4.01]	0.88 [3.96]	0.91 [3.84]	0.22 [2.49]	0.23 [2.63]	0.21 [2.40]	0.25 [2.79]	0.24 [2.64]	0.28 [2.97]
	(3)	0.63 [2.90]	0.74 [3.55]	0.76 [3.83]	0.84 [4.26]	0.89 [4.11]	0.27 [3.31]	0.27 [3.33]	0.24 [2.92]	0.21 [2.57]	0.20 [2.43]	0.18 [2.22]
	(4)	0.63 [3.11]	0.65 [3.50]	0.70 [3.80]	0.71 [3.87]	0.85 [4.15]	0.22 [3.02]	0.22 [2.96]	0.17 [2.37]	0.15 [2.04]	0.15 [2.05]	0.14 [1.83]
	(5)	0.42 [2.25]	0.50 [2.91]	0.56 [3.75]	0.66 [4.26]	0.63 [3.63]	0.22 [2.45]	0.26 [3.00]	0.25 [2.85]	0.24 [2.70]	0.29 [3.30]	0.29 [3.15]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CEI Quintiles					CEI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	258	259	260	256	254	24	24	22	21	22	
	(2)	79	79	79	79	78	42	42	43	42	43	
	(3)	61	61	61	60	60	78	78	77	78	78	
	(4)	55	55	55	55	55	182	177	181	178	177	
(5)	52	52	52	52	51	1116	1324	1860	1507	1258		

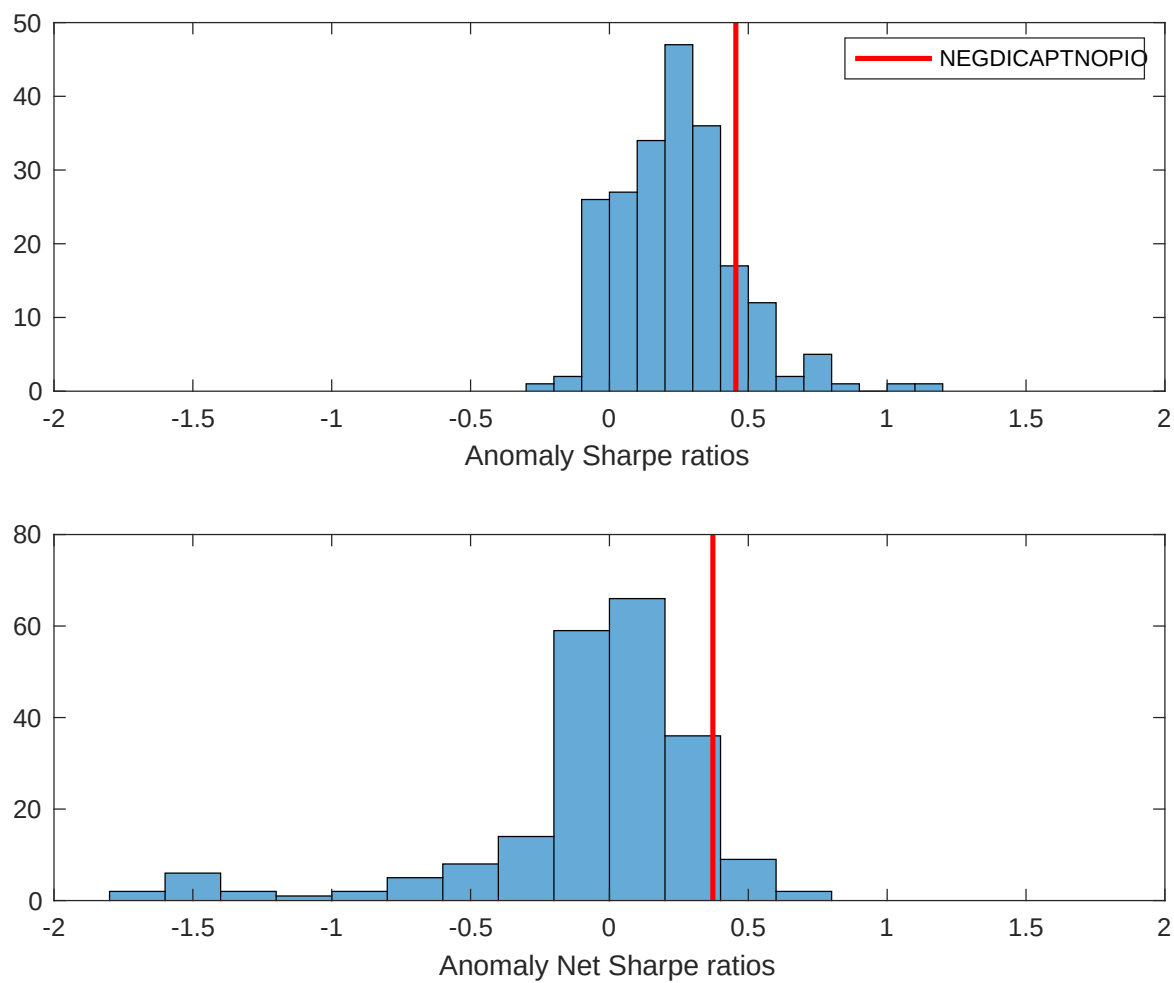


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

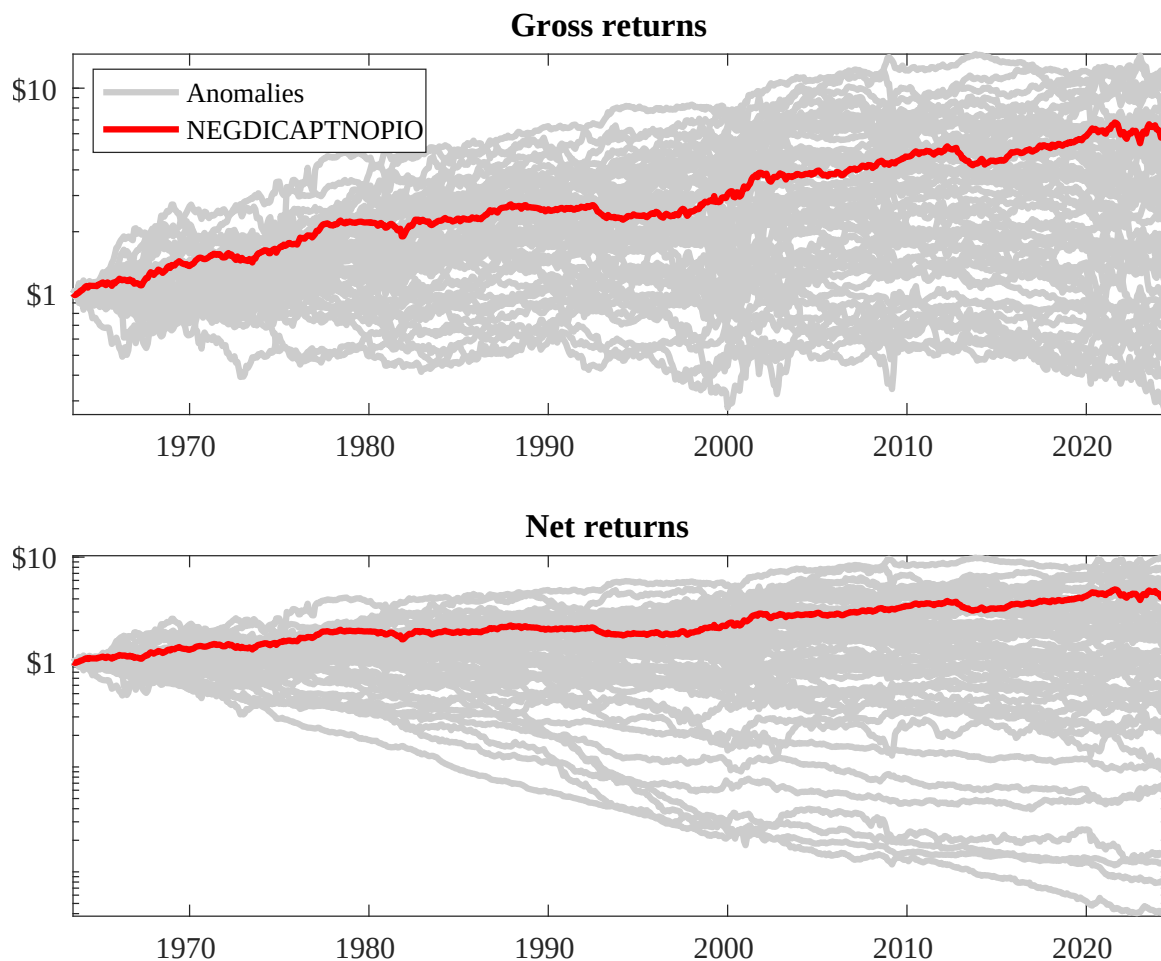
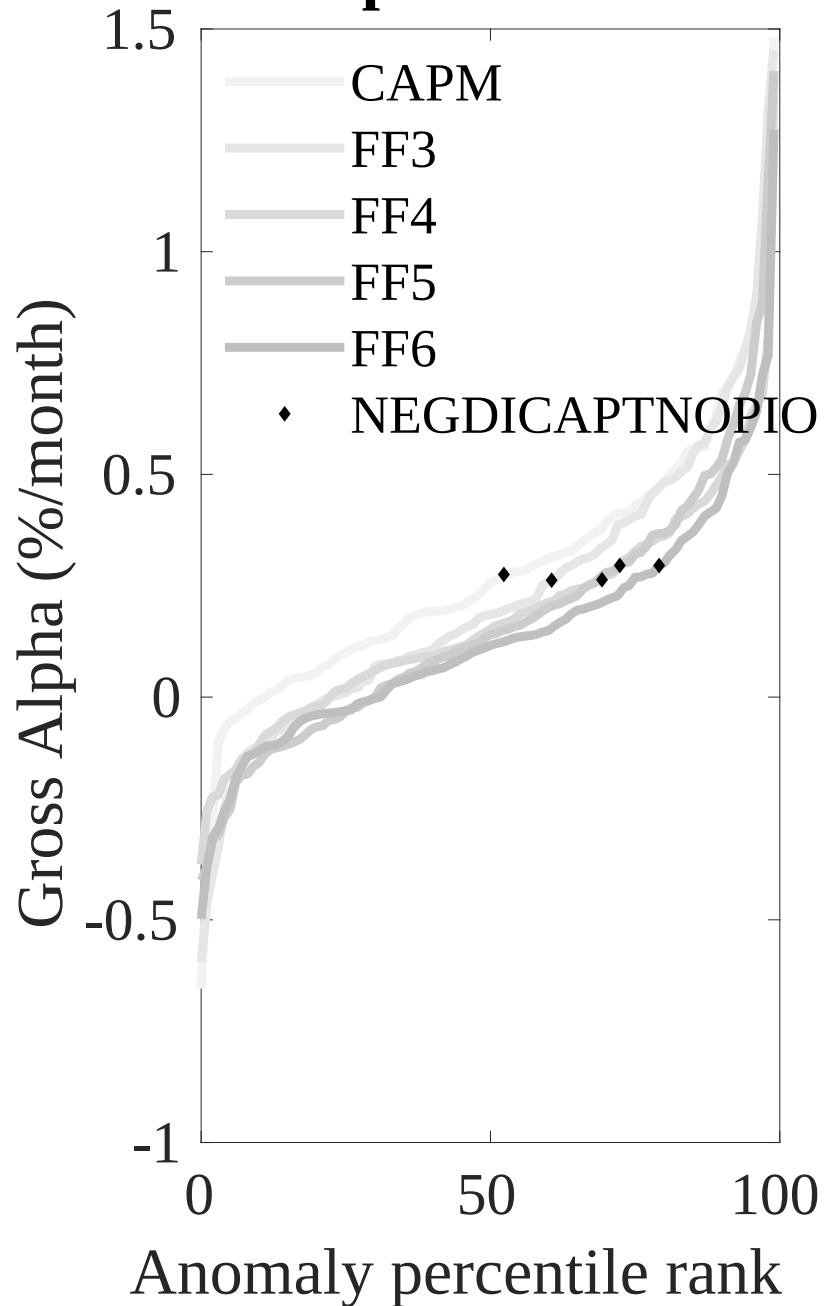


Figure 3: Dollar invested.

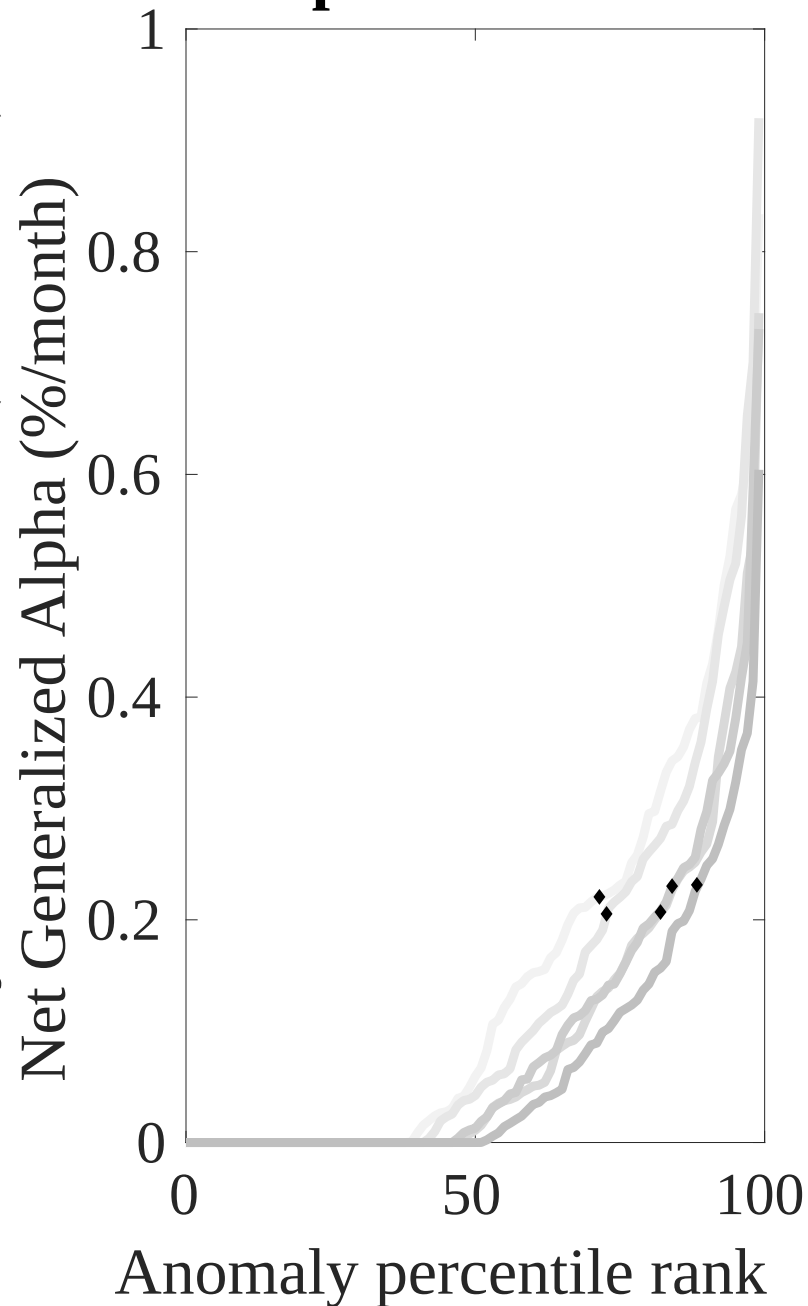
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



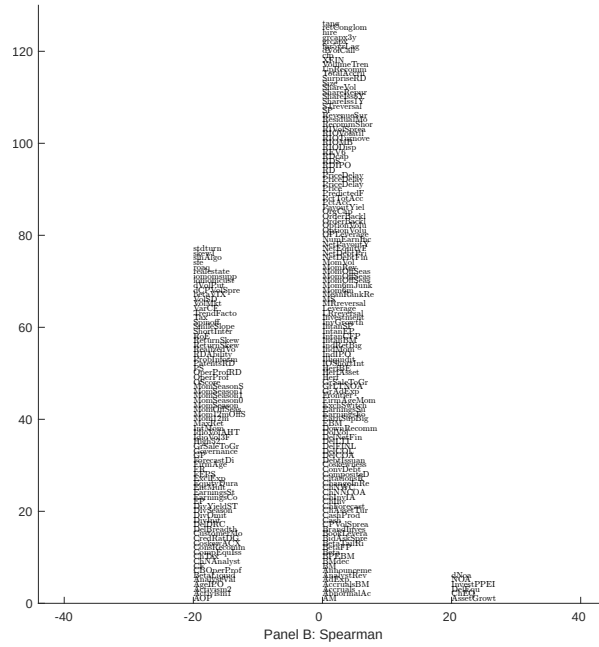
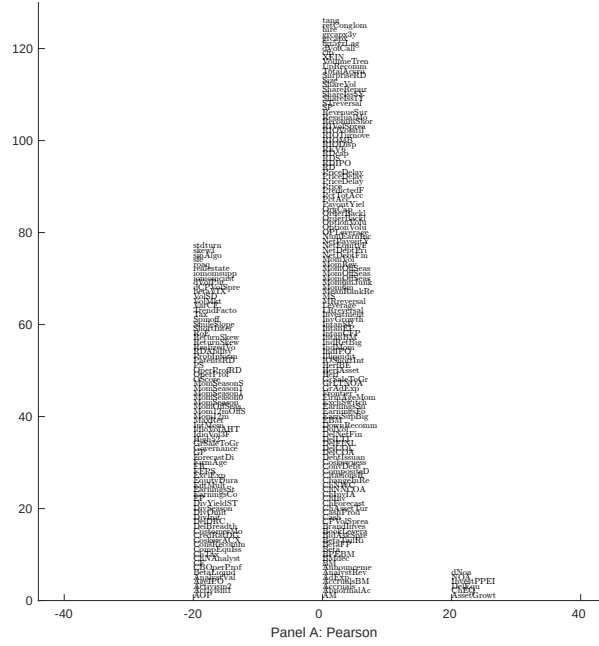


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

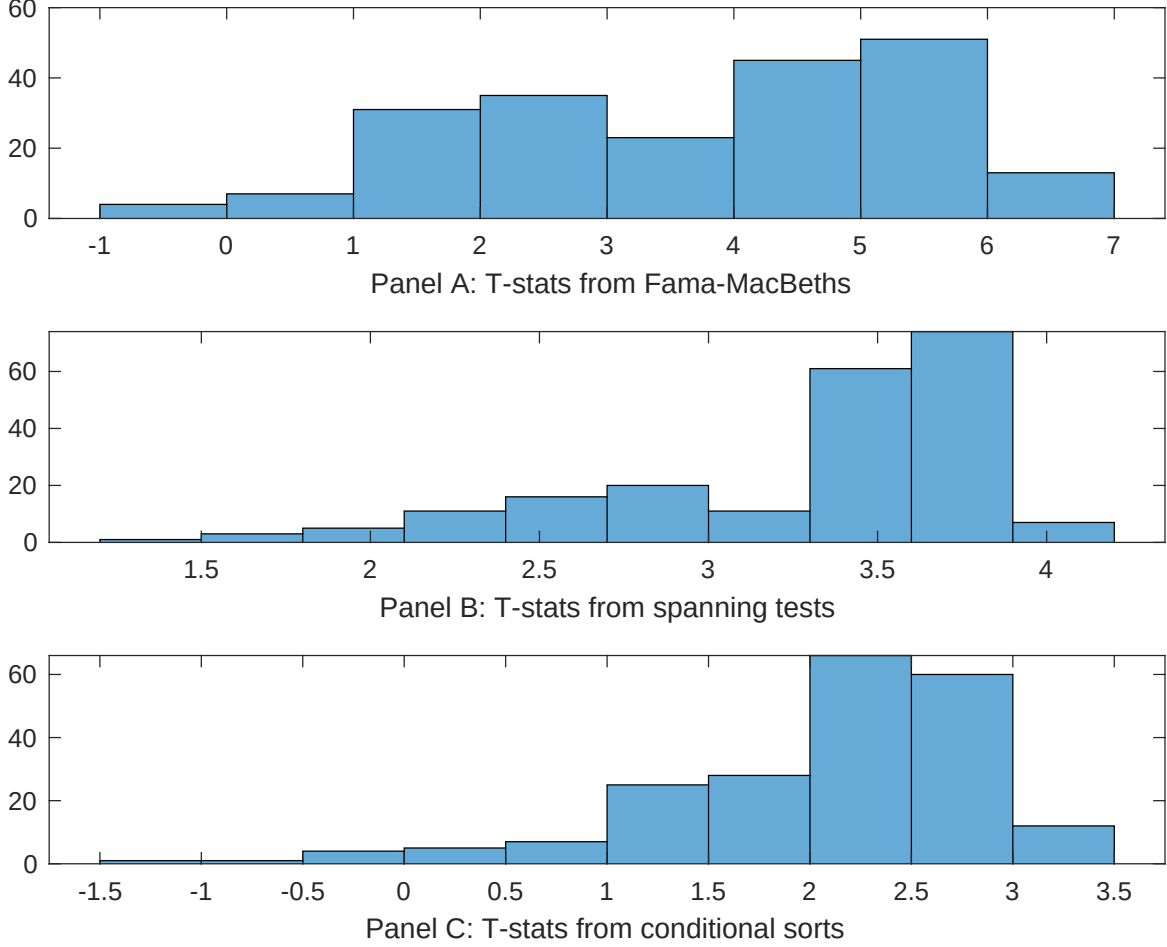


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEI}CEI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [6.82]	0.13 [6.03]	0.12 [5.58]	0.13 [5.67]	0.13 [5.82]	0.17 [7.02]	0.15 [5.03]
CEI	0.37 [4.24]	0.24 [2.89]	0.41 [4.56]	0.34 [3.22]	0.16 [1.86]	0.29 [3.46]	0.95 [0.93]
Anomaly 1	0.41 [3.57]						0.81 [0.72]
Anomaly 2		0.16 [7.83]					0.43 [1.43]
Anomaly 3			0.29 [5.58]				0.28 [0.41]
Anomaly 4				0.49 [8.13]			0.94 [1.30]
Anomaly 5					0.13 [9.13]		0.61 [2.43]
Anomaly 6						0.83 [6.85]	0.19 [1.08]
# months	720	732	732	732	720	732	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28 [3.84]	0.29 [3.93]	0.26 [3.59]	0.29 [3.91]	0.28 [3.80]	0.26 [3.57]	0.24 [3.34]
Anomaly 1	12.12 [3.06]						10.23 [2.53]
Anomaly 2		7.91 [2.31]					2.73 [0.70]
Anomaly 3			12.85 [3.57]				12.55 [2.76]
Anomaly 4				6.09 [2.06]			-2.53 [-0.67]
Anomaly 5					3.80 [0.92]		-5.74 [-1.26]
Anomaly 6						9.68 [3.14]	8.90 [2.70]
mkt	-3.43 [-1.96]	-3.93 [-2.25]	-3.42 [-1.96]	-3.91 [-2.24]	-3.81 [-2.17]	-4.18 [-2.39]	-3.45 [-1.98]
smb	-1.15 [-0.45]	-1.06 [-0.42]	0.81 [0.32]	-0.34 [-0.13]	-0.70 [-0.28]	0.80 [0.31]	1.51 [0.58]
hml	-0.20 [-0.06]	0.43 [0.13]	0.44 [0.13]	1.00 [0.29]	1.00 [0.30]	-1.32 [-0.38]	-2.94 [-0.85]
rmw	-12.84 [-3.77]	-13.46 [-3.95]	-11.22 [-3.26]	-12.70 [-3.70]	-13.39 [-3.91]	-14.43 [-4.23]	-12.08 [-3.50]
cma	-5.11 [-0.82]	0.56 [0.10]	-1.78 [-0.32]	1.64 [0.30]	3.80 [0.65]	6.48 [1.31]	-7.38 [-1.06]
umd	-0.19 [-0.11]	0.01 [0.01]	-0.75 [-0.43]	-0.63 [-0.35]	-0.05 [-0.03]	-0.32 [-0.19]	-0.86 [-0.49]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	4	4	5	4	3	4	6

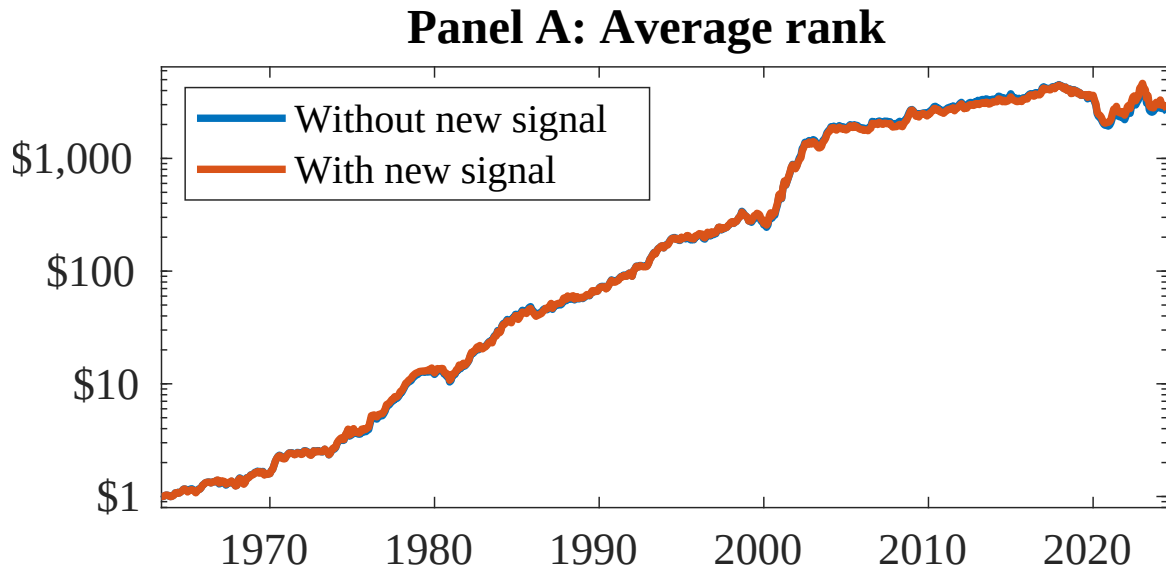


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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